SOUTH LAKE HOME EDUCATORS



Lesson 3

Programming Basics II: Take Your Shot

Semester 1, Week 3

At a Glance

Lesson 3 objectives & materials

By the end of today, students will:

- Explain the template/instance relationship
- Explain why spawned objects must be cleaned up
- Explain "just pressed" vs. "currently held" input checks
- Add a working shoot mechanic to their Semester 1 project

What We're Using Today

- Demo laptop with all 8 engines verified
- Godot, for the live lecture demo
- Engine-specific Student Handouts, one per student
- This lesson's Assets folder (new projectile sprite + sound - no download needed)
- Your own [YourName]Game project from Lesson 2 - not a new one

IGDA framework tie-in: Programming - object instancing, resource/memory management, input handling (introductory).

Today's Agenda



0:00 - 0:05	Welcome back, Lesson 2 recap, quick share-out
0:05 - 0:25	Instances: One Template, Many Copies (live demo in Godot)
0:25 - 0:45	Cleanup: Why You Have to Remove What You Create
0:45 - 0:50	One Press, One Shot (edge-triggered vs. held-down input)
0:50 - 1:00	Introduce lab exercise, hand out engine-specific handouts
LAB TIME	***— Take Your Shot —***
1:00 - 1:50	Add shooting to your ongoing Semester 1 project
1:50 - 2:00	Exit check: quick share-out



Welcome Back

Lesson 2 recap and a quick share of any finished Key Press Reactors.

5 min

Welcome Back

5 min

Quick recap of Lesson 2:

- Sequences, events, and the game loop - and the start of your Semester 1 project
- Everyone built a Key Press Reactor moving a character in up to 4 directions
- Today: adding shooting - the first piece of your capstone's actual mechanic

Open your own project from last week today, not a new one - everything we build adds on top of the character you already have.



Instances: One Template, Many Copies

How pressing fire five times gives you five separate, independent bullets.

20 min

Instances

20 min

Build ONE definition of a bullet. Every press stamps out a brand-new, independent COPY of it - that's an instance.

Change one bullet's position, and none of the others are affected - they're all separate, all remembering their own data.

Same idea, nearly the same word, across every engine:

Godot	`bullet_scene.instantiate()`
GameMaker	`instance_create_layer()`
Unity	`Instantiate()`
Unreal	`Spawn Actor from Class`



Discussion: where else in a game might you want lots of independent copies of one thing?



Cleanup: Why You Remove What You Create

A hundred forgotten bullets doesn't sound like much - until it does.

20 min

Cleanup

20 min

If nothing removes old bullets, the engine keeps tracking every single one - forever - even off-screen and useless.

A real game spawning constantly (bullets, particles, enemies) can pile up thousands of forgotten objects in minutes - the game slows down or crashes.

Every engine gives you an explicit "I'm done with this one" call:

Godot	`queue_free()`
GameMaker	`instance_destroy()`
Unity / Unreal	`Destroy()`



The pattern for anything spawned at runtime is always two-sided: something that creates it, and something that eventually destroys it. Today, a bullet leaving the screen triggers that second half.



One Press, One Shot

"Currently held" vs. "just pressed" - the difference that keeps fire from flooding the screen.

5 min

One Press, One Shot

5 min

Two different questions you can ask about a key:

Currently held?

True for as long as the key is down - checked fresh every frame. Right for movement: hold, keep moving.

Just pressed?

True for exactly one frame, no matter how long you hold it after. Right for firing: one press, one shot.

Same split, different names, in nearly every engine:

Godot	`is_action_just_pressed()`	vs.	`is_action_pressed()`
GameMaker	`keyboard_check_pressed()`	vs.	`keyboard_check()`
Unity	`wasPressedThisFrame`	vs.	`isPressed`



Today's Exercise & Lab

Take Your Shot - the first piece of your capstone's actual shooting mechanic.

10 min

Take Your Shot

10 min

Pressing your fire key spawns a projectile that flies upward and disappears off-screen - added on top of your existing project.

Level 1 - Minimal

One press spawns one projectile that flies up and disappears.

MVP - a complete finish for today.

Level 2 - Complete

Repeatable, multiple projectiles, capped by a fire-rate cooldown.

Everything works end to end.

Level 3 - Refined

Sound, a distinct look, and/or clean code structure - pick what interests you.

Ready to share.



Reminder: if your Lesson 2 character still moves in all 4 directions, that's fine to keep - or trim it to Left/Right to match the actual capstone game exactly. Either way works.

Lab Time Notes



- Most common bug this week: a held-down fire key spawning dozens of bullets per second - check for a "held" input check where a "just pressed" one belongs.
- Second most common: bullets that spawn but never move or never disappear - usually the movement/cleanup code is in the wrong event.
- Students who finish Level 2 early should be pointed at the Level 3 ideas on their handout - not given new unrelated work.
- At 1:50, call time and do a 2-minute share-out: ask 2-3 students to show what they built, across different engines if possible.

Great Work Today!

Next week: a single enemy on screen, and collision detection - your projectile finally has something to hit.

Keep polishing your shooting mechanic at home if you'd like - try one of the Level 3 ideas if you haven't already.